Multi-Armed Bandits: Exploration, Exploitation, and Action-Value Methods

Saif Ur Rehman

March 2026

Abstract

This report investigates the multi-armed bandit problem using a set of experiments based on the framework introduced by Sutton and Barto. Problem 1 implements a stationary k-armed bandit and analyses the exploration–exploitation trade-off using greedy and ϵ-greedy policies. Problem 2 compares several exploration strategies, including ϵ-greedy, optimistic initial values, upper confidence bound (UCB), softmax exploration, and gradient bandits, together with a parameter sensitivity study. Problem 3 examines action-value estimation methods in both stationary and nonstationary environments, comparing sample-average, constant step-size, unbiased constant step-size, and sliding-window estimators. The results show that exploration is essential for strong long-term performance, that UCB provides the strongest exploration strategy in the stationary testbed, and that adaptive estimators such as unbiased constant step-size and sliding-window methods are more effective in nonstationary environments.

Group Members and Contributions

Saif Ur Rehman
Email: sgsrehm1@liverpool.ac.uk
Student ID: 201940185

Contribution:

This assignment was completed individually. The author was responsible for the implementation of all algorithms, experimental design, generation of experimental results, analysis of findings, and preparation of the final report.

Introduction

The multi-armed bandit problem is a classical reinforcement learning problem in which an agent repeatedly selects from a set of actions in order to maximise cumulative reward over time [1, 4]. Each action has an unknown reward distribution, and the agent must decide whether to exploit the action that currently appears best or to explore alternative actions that may prove more rewarding in the long run. This exploration–exploitation trade-off is one of the central challenges in reinforcement learning.

Following Sutton and Barto, the experiments in this report use Gaussian bandit testbeds in which the true value of each action is sampled from a normal distribution and observed rewards are noisy samples centred around those action values. To obtain reliable estimates of algorithm performance, results are averaged over many independent runs rather than relying on the outcome of a single bandit instance.

In the bandit setting, the objective is typically to maximise the expected cumulative reward over a sequence of action selections. This objective can also be expressed in terms of minimising cumulative regret, which measures the difference between the reward obtained by the learning algorithm and the reward that would have been obtained by always selecting the optimal action.

Designing algorithms that balance exploration and exploitation efficiently is therefore central to achieving strong performance in bandit problems. The experiments in this report investigate how different exploration strategies and action-value estimation methods influence learning behaviour under both stationary and nonstationary reward distributions.

Methodology

For stationary experiments, the true action values were sampled as

$$q^*(a) \sim \mathcal{N}(0, 1),$$

and rewards were generated as

$$R_t \sim \mathcal{N}(q^*(A_t), 1).$$

For nonstationary experiments, the true action values were additionally updated by a Gaussian random walk so that the optimal action could change over time. A fixed random seed was used to ensure reproducibility. Within each run, the same underlying bandit instance was shared across algorithms, while each agent used independent random number streams for its action selection and reward sampling. Random tie-breaking was used whenever multiple actions had the same estimated value.

Performance was evaluated using:

- average reward over time,
- percentage of optimal actions selected,
- final performance at the last time step,
- normalised area under the curve (AUC), computed as the mean over time.

To improve the readability of the learning curves, a moving-average smoothing window of size 50 was applied when plotting the reward and optimal-action curves. This window size provides sufficient noise reduction while preserving the overall learning dynamics over the 2000-step horizon. The smoothing was applied only for visualisation of the plots; all reported statistics and summary tables were computed using the raw experimental data.

1 Implementation of a Multi-Armed Bandit Algorithm

1.1 Setup

1.1.1 Environment

Problem 1 considers a stationary 10-armed bandit evaluated over 2000 action selections. The stationary bandit environment follows the standard Gaussian testbed described by Sutton and Barto. The true action values are sampled as

$$q^*(a) \sim \mathcal{N}(0, 1),$$

and the reward observed after selecting action A_t is sampled as

$$R_t \sim \mathcal{N}(q^*(A_t), 1).$$

This formulation ensures that rewards are stochastic observations around the underlying action values.

All results were averaged across 2000 independent runs in order to obtain stable performance estimates and reduce variance caused by stochastic rewards.

The objective of the k-armed bandit problem is to maximise the expected cumulative reward obtained over a sequence of action selections. Since the true action values are unknown, the agent must learn from observed rewards while balancing exploration and exploitation. Policies that discover the optimal action earlier achieve higher cumulative reward and therefore incur lower regret over time.

1.1.2 Exploration Policies

To investigate the exploration–exploitation trade-off, three simple policies were evaluated:

- greedy ($\epsilon = 0$),
- ϵ-greedy ($\epsilon = 0.01$),
- ϵ-greedy ($\epsilon = 0.1$).

The greedy policy always selects the action with the highest estimated value, representing pure exploitation. In contrast, the ϵ-greedy strategy selects the best estimated action with probability $1 - \epsilon$ and explores a random action with probability ϵ. This simple mechanism allows the agent to continue sampling alternative actions while primarily exploiting the best known option.

1.1.3 Action-Value Estimation

Action values were estimated using the sample-average update rule

$$Q_{t+1}(a) = Q_t(a) + \frac{1}{N_t(a)}\big(R_t - Q_t(a)\big),$$

where $N_t(a)$ denotes the number of times action a has been selected up to time t.

This estimator computes the empirical mean of observed rewards for each action and is an unbiased estimator of the true action value in stationary environments [1, 5].

The purpose of this experiment is to demonstrate the fundamental exploration–exploitation dilemma. Greedy action selection corresponds to pure exploitation, while ϵ-greedy introduces controlled random exploration.

1.2 Results

Figure 1 shows the average reward obtained per time step, averaged across 2000 independent runs. The greedy strategy improves quickly during the early stages but soon stabilises at a relatively low level. This behaviour occurs because the agent may commit prematurely to a suboptimal action due to noisy initial rewards. Once a greedy policy converges to an incorrect action estimate, it has no mechanism to recover.

Both ϵ-greedy strategies perform better because they continue to sample alternative actions. The policy with $\epsilon = 0.1$ learns faster in the early stage due to stronger exploration, while $\epsilon = 0.01$ gradually achieves higher average reward as the number of steps increases because it performs less random exploration once good value estimates have been learned.

Figure 2 shows the percentage of optimal action selections. The greedy method plateaus near 35%, demonstrating that pure exploitation is insufficient for identifying the optimal arm. In contrast, ϵ-greedy policies continue improving throughout the experiment because they repeatedly explore alternative actions. The stronger exploratory policy, $\epsilon = 0.1$, achieves the highest optimal-action rate, ending at approximately 84%.

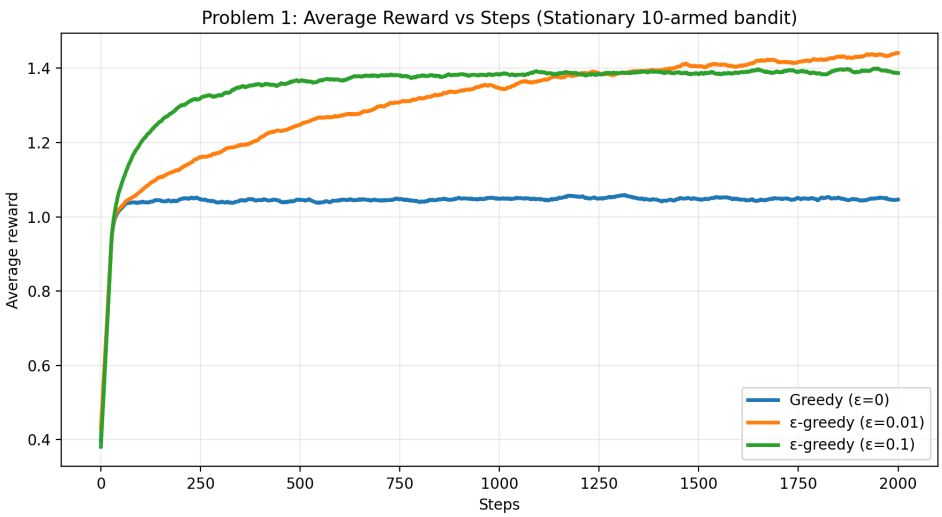

Figure 1: Problem 1: Average reward over time for greedy and ϵ-greedy policies in the stationary 10-armed bandit.

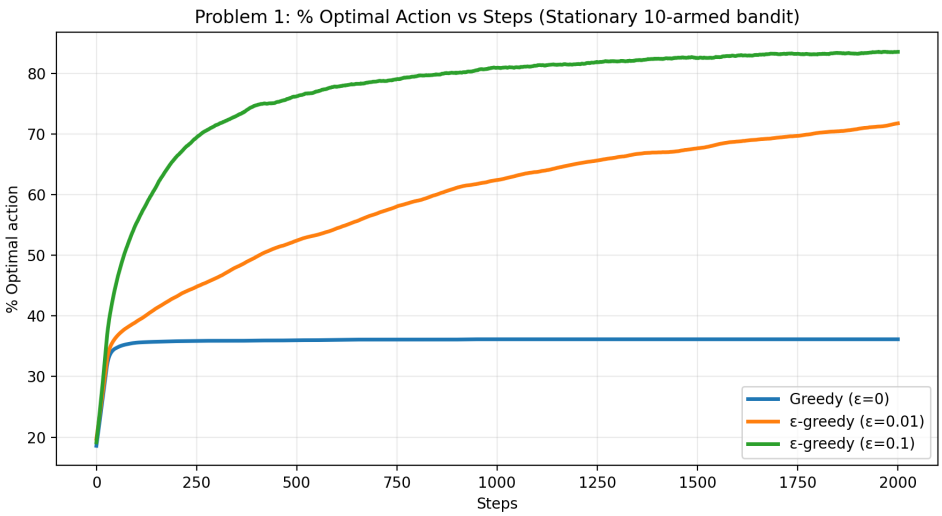

Figure 2: Problem 1: Percentage of optimal action selections over time for greedy and ϵ-greedy policies.

1.3 Sensitivity to the Number of Arms

Additional experiments were performed for $n = 5$, $n = 10$, and $n = 20$ arms in order to examine how exploration behaviour changes as the action space grows. Table 1 summarises the best and runner-up policies according to the area-under-curve (AUC) average reward.

Table 1: Sensitivity of exploration strategies to the number of arms.

n	Best Policy	Runner-Up	Final Avg	AUC Avg	Std Avg	Final % Opt	AUC % Opt	Std % Opt
5	ϵ-greedy (0.01)	ϵ-greedy (0.1)	1.1101	1.0480	0.0741	84.70	75.85	7.7
10	ϵ-greedy (0.1)	ϵ-greedy (0.01)	1.3908	1.3538	0.0967	83.60	77.10	9.78
20	ϵ-greedy (0.1)	ϵ-greedy (0.01)	1.6581	1.6024	0.1523	78.65	67.54	14.21

The results show that the preferred exploration rate depends on the size of the action space.

With only five arms, a smaller exploration rate is sufficient because the agent can identify the best action relatively quickly. As the number of arms increases, stronger exploration becomes more beneficial because the agent must search a larger action space before identifying the optimal action.

1.4 Discussion

Problem 1 clearly demonstrates the exploration–exploitation trade-off. A purely greedy strategy performs poorly because it has no mechanism to recover from incorrect early estimates caused by stochastic rewards. ϵ-greedy policies overcome this limitation by maintaining a small probability of exploring alternative actions.

At the same time, the results illustrate the cost of excessive exploration. Stronger exploration improves the likelihood of identifying the optimal arm, but a lower exploration rate may achieve slightly higher reward once accurate value estimates have been learned. This behaviour can also be interpreted through the concept of regret minimisation, which measures the performance loss relative to always selecting the optimal action [3]. Policies that discover the optimal action earlier therefore incur lower cumulative regret.

These results motivate the more advanced exploration strategies examined in Problem 2, which aim to balance exploration and exploitation more efficiently than simple ϵ-greedy policies.

2 Exploration and Exploitation in Multi-Armed Bandits

2.1 Setup

2.1.1 Environment

The experiments in this section use the stationary 10-armed bandit testbed introduced by Sutton and Barto. This configuration is commonly used for comparing exploration strategies because the underlying reward distributions remain fixed, allowing differences in performance to be attributed directly to the action-selection policy rather than changes in the environment.

Using a fixed 10-armed testbed also enables direct comparison with the results presented in the reference literature.

Nonstationary environments are not considered in this experiment because the objective of Problem 2 is to analyse exploration mechanisms rather than adaptation to changing reward distributions. The impact of nonstationarity on learning behaviour is instead examined in Problem 3, where different action-value estimation methods are evaluated in both stationary and nonstationary settings.

2.1.2 Exploration Strategies

Five exploration strategies were evaluated in the stationary 10-armed bandit testbed:

- ϵ-greedy ($\epsilon = 0.1$),
- optimistic initial values ($Q_0 = 5$, $\alpha = 0.1$),
- upper confidence bound (UCB) ($c = 1$),
- softmax / Boltzmann exploration ($\tau = 0.1$),
- gradient bandit ($\alpha = 0.1$).

Each algorithm represents a different approach for addressing the exploration–exploitation trade-off while interacting with a stationary Gaussian bandit environment.

1. ϵ-Greedy The ϵ-greedy policy selects the action with the highest estimated value most of the time, but with probability ϵ it chooses an action uniformly at random.

$$A_t = \begin{cases} \text{random action} & \text{with probability } \epsilon \\ \arg\max_a Q_t(a) & \text{with probability } 1-\epsilon \end{cases}$$

In this experiment $\epsilon = 0.1$, meaning that the agent explores randomly in 10% of the action selections.

2. Optimistic Initial Values Optimistic initial values encourage exploration by initializing action-value estimates with a large value:

$$Q_0(a) = Q_{\text{init}}.$$

In the experiments we use $Q_{\text{init}} = 5$. Because all actions initially appear promising, the agent naturally explores each action before converging to the best one.

Action values are updated using a constant step-size rule

$$Q_{t+1}(a) = Q_t(a) + \alpha\big(R_t - Q_t(a)\big),$$

where $\alpha = 0.1$ controls the learning rate.

3. Upper Confidence Bound (UCB) The UCB algorithm selects actions according to

$$A_t = \arg\max_a \left[Q_t(a) + c\sqrt{\frac{\ln t}{N_t(a)}} \right].$$

This approach balances exploration and exploitation by assigning an exploration bonus to actions with greater uncertainty [2].

The second term is an uncertainty bonus that decreases as an action is sampled more frequently. In this experiment $c = 1$, balancing exploration and exploitation. Actions that have been selected fewer times receive a larger bonus, encouraging systematic exploration.

4. Softmax / Boltzmann Exploration Softmax exploration chooses actions probabilistically according to their estimated values:

$$\pi_t(a) = \frac{e^{Q_t(a)/\tau}}{\sum_b e^{Q_t(b)/\tau}}.$$

The temperature parameter τ controls the level of exploration. Large values produce nearly uniform exploration, while small values produce behaviour closer to greedy action selection. In the experiments $\tau = 0.1$ was used.

5. Gradient Bandit The gradient bandit algorithm learns preferences $H_t(a)$ for each action rather than estimating action values directly. The policy is defined using a softmax distribution:

$$\pi_t(a) = \frac{e^{H_t(a)}}{\sum_b e^{H_t(b)}}.$$

Preferences are updated using stochastic gradient ascent:

$$H_{t+1}(a) = H_t(a) + \alpha(R_t - \bar{R}_t)(\mathbf{1}_{a=A_t} - \pi_t(a)),$$

where $\bar{R}_t$ is the average reward baseline and $\alpha = 0.1$ is the step size. The baseline helps reduce variance in the updates and stabilizes learning.

The environment and reward model were kept fixed so that performance differences can be attributed to the exploration mechanism rather than changes in bandit difficulty.

2.2 Results

Figure 3 compares average reward over time. UCB provides the strongest overall performance, achieving the highest reward for most of the horizon. This reflects the fact that UCB uses an explicit uncertainty bonus and therefore explores more efficiently than purely random exploration methods. Optimistic initial values and gradient bandit also perform strongly and approach similar long-run reward levels. Optimistic initial values learn more slowly in the early stage because the high initial estimates encourage the agent to sample each action before converging toward the best one. In contrast, ϵ-greedy remains weaker because its exploration is purely random, while softmax performs worst among the methods considered here.

Figure 4 compares the percentage of optimal action selections. UCB reaches the highest optimal-action rate because its confidence-bound term systematically encourages exploration of actions with uncertain value estimates. Optimistic initial values and gradient bandit also perform strongly, while the ϵ-greedy strategy performs moderately well but remains below these methods. Softmax is considerably weaker in this configuration because it continues assigning probability to suboptimal actions even after their value estimates become worse. These results suggest that uncertainty-driven and preference-based exploration are more effective in this stationary bandit testbed than purely probabilistic value-based exploration.

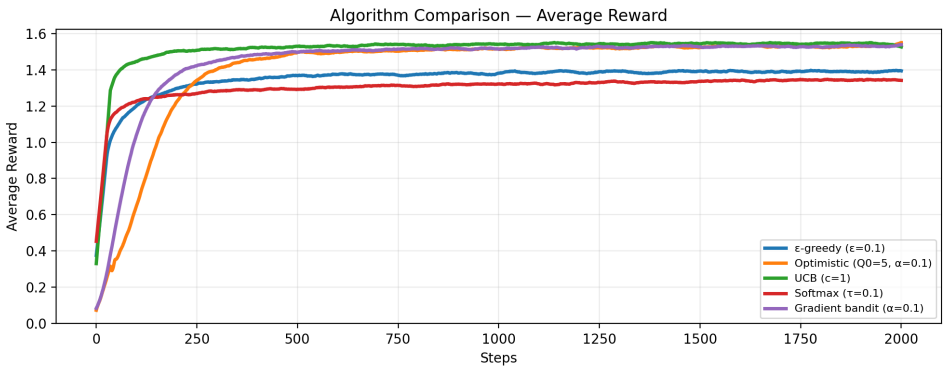

Figure 3: Problem 2: Average reward comparison of exploration strategies in the stationary 10-armed bandit.

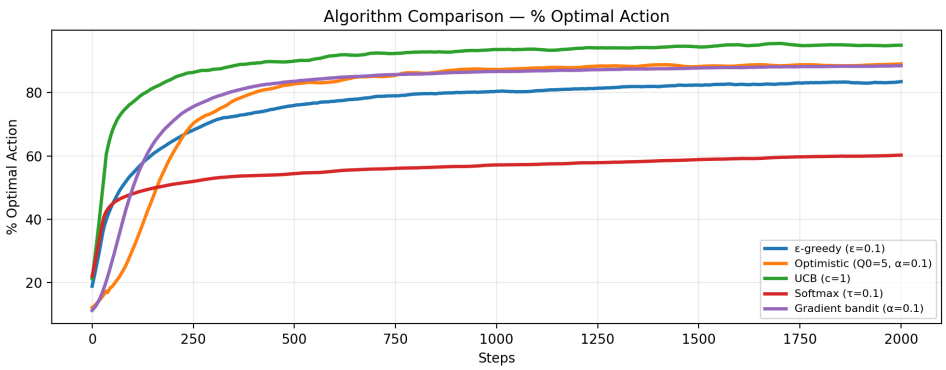

Figure 4: Problem 2: Percentage of optimal action selections for the exploration strategies.

2.3 Parameter Sensitivity

Figure 5 shows how average reward changes as the key exploration parameter is varied for each method. In general, performance is highest at intermediate parameter values, indicating that both insufficient and excessive exploration reduce long-run reward. Very small parameter values lead to insufficient exploration, while very large values cause excessive exploration and reduce the ability of the agent to exploit high-reward actions. Based on these experiments, the best-performing parameters were:

- ϵ-greedy: $\epsilon = 0.03125$,
- UCB: $c = 1$,
- optimistic values: $Q_0 = 1$,
- softmax: $\tau = 0.25$,
- gradient bandit: $\alpha = 0.5$.

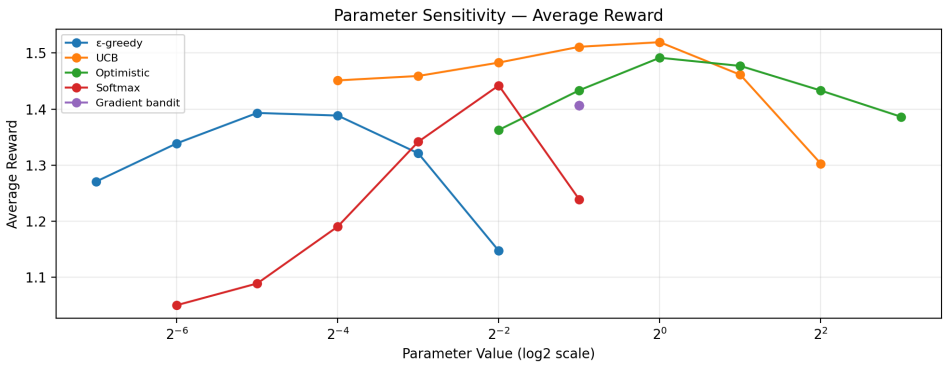

Figure 5: Problem 2: Parameter sensitivity study for the exploration strategies.

Table 2: Problem 2A: Best-performing parameter for each exploration strategy.

Agent	Best Parameter	Best Avg Reward
ϵ-greedy	0.03125	1.3931
UCB	1.0	1.5192
Optimistic values	1.0	1.4915
Softmax	0.25	1.4421
Gradient bandit	0.5	1.4066

2.4 Scaling with the Number of Arms

An additional scaling study was performed for $n = 5, 10, 20$ arms using the exploration strategies introduced earlier. Table 3 summarises the best-performing method and key performance metrics for each setting. Across all three settings, UCB achieved the highest AUC average reward, confirming that uncertainty-driven exploration scales better than the alternatives in this experiment. Gradient bandit consistently achieved the second-best performance across the tested settings.

As the number of arms increases, the percentage of optimal actions decreases slightly, reflecting the increased difficulty of identifying the optimal arm in larger action spaces. Despite

this increase in problem complexity, UCB remains the most robust method among the strategies considered.

Table 3: Problem 2B: Scaling summary across different numbers of arms.

n	Best Agent	Runner-Up	Final Avg	AUC Avg	Final % Opt	AUC % Opt
5	UCB ($c = 1$)	Gradient bandit ($\alpha = 0.1$)	1.1647	1.1607	96.30	92.87
10	UCB ($c = 1$)	Gradient bandit ($\alpha = 0.1$)	1.5088	1.5045	94.90	90.58
20	UCB ($c = 1$)	Gradient bandit ($\alpha = 0.1$)	1.8843	1.8346	94.75	87.27

2.5 Discussion

Problem 2 shows that exploration strategy strongly influences learning performance. Random exploration through ϵ-greedy improves over purely greedy selection but remains less efficient than directed exploration strategies. In particular, UCB performs best in the stationary bandit testbed because its confidence-bound term explicitly encourages exploration of actions with high uncertainty. Optimistic initial values promote exploration during the early stages of learning by assigning high initial estimates to all actions, encouraging the agent to sample each arm before converging toward the most promising one. Gradient bandits also perform competitively by learning action preferences through a stochastic policy rather than relying on explicit value estimates. In contrast, softmax exploration is less effective in this setting because it continues assigning probability to weaker actions even after their estimated values become worse. The scaling experiment further shows that UCB remains the strongest strategy as the number of arms increases, indicating that uncertainty-driven exploration remains effective even as the search space grows. Overall, the results demonstrate that the effectiveness of exploration depends strongly on how exploratory actions are selected, with directed exploration mechanisms providing more efficient learning than purely random exploration.

3 Action-Value Methods

3.1 Setup

3.1.1 Environment

In contrast to Problem 2, where the environment was stationary, this experiment evaluates action-value estimators in both stationary and nonstationary bandit environments. The stationary environment follows the standard Gaussian bandit testbed, where the true action values are sampled as

$$q^*(a) \sim \mathcal{N}(0, 1).$$

To evaluate the ability of estimators to adapt to changing reward distributions, a nonstationary bandit environment is also considered. In the nonstationary setting, the true action values evolve over time according to a Gaussian random walk

$$q^*_{t+1}(a) = q^*_t(a) + \mathcal{N}(0, \sigma^2).$$

This causes the optimal action to change over time and therefore requires the learning algorithm to continuously update its value estimates.

3.1.2 Exploration Policy

In order to isolate the effect of the value estimation method, all algorithms in this experiment use the same action-selection policy. An ϵ-greedy policy with $\epsilon = 0.1$ is used for all estimators.

This ensures that differences in performance arise from the value estimation method rather than from differences in exploration behaviour.

3.1.3 Action-Value Estimation Methods

Problem 3 compares several action-value estimation methods under a common ϵ-greedy exploration policy ($\epsilon = 0.1$). The estimators differ in how they update the action-value estimate $Q_t(a)$ when a reward is observed.

The estimators compared are:

- sample-average estimator,
- constant step-size estimator ($\alpha = 0.1$),
- unbiased constant step-size estimator ($\alpha = 0.1$),
- sliding-window estimator ($W = 50$).

All methods update the estimate of the selected action after observing reward R_t at time step t. However, the way past rewards are incorporated differs between estimators.

1. Sample-Average Estimator The sample-average estimator updates the action value using the mean of all observed rewards for that action:

$$Q_{t+1}(a) = Q_t(a) + \frac{1}{N_t(a)} \left(R_t - Q_t(a)\right),$$

where $N_t(a)$ is the number of times action a has been selected up to time t.

This estimator gives equal weight to all past observations. As a result, it is an unbiased estimator when the environment is stationary. However, its effective learning rate decreases as $N_t(a)$ grows, making it slow to adapt when the underlying reward distribution changes in nonstationary environments.

2. Constant Step-Size Estimator The constant step-size estimator uses a fixed learning rate α:

$$Q_{t+1}(a) = Q_t(a) + \alpha \left(R_t - Q_t(a)\right).$$

In this study we use $\alpha = 0.1$. Because the learning rate does not decrease over time, recent rewards have greater influence than older observations. This property makes the estimator more suitable for nonstationary environments where action values may drift over time.

3. Unbiased Constant Step-Size Estimator The constant step-size estimator introduces bias during early learning because initial value estimates influence the updates. To reduce this effect, an unbiased step-size can be used by introducing a normalisation term:

$$\beta_t = \frac{\alpha}{o_t},$$

where

$$o_{t+1} = o_t + \alpha(1 - o_t).$$

The update rule then becomes

$$Q_{t+1}(a) = Q_t(a) + \beta_t \left(R_t - Q_t(a)\right).$$

This formulation preserves the adaptability of constant step-size updates while reducing the bias caused by the initial value estimates.

4. Sliding-Window Estimator The sliding-window estimator maintains an average over only the most recent W observations:

$$Q_t(a) = \frac{1}{W} \sum_{i=t-W+1}^{t} R_i.$$

In this experiment a window size of $W = 50$ was used. By discarding older rewards, the estimator focuses on recent observations and can therefore respond quickly when the true action values change in a nonstationary environment. However, if the window size is too small, the estimates may become noisy.

Stationary vs Nonstationary Environments In the stationary experiments, the true action values remain fixed throughout the interaction:

$$q^*(a) \sim \mathcal{N}(0, 1).$$

In the nonstationary experiments, the true action values evolve according to a Gaussian random walk:

$$q^*_{t+1}(a) = q^*_t(a) + \mathcal{N}(0, \sigma^2).$$

This drift causes the optimal action to change over time. Consequently, estimators that place more weight on recent observations are expected to perform better in the nonstationary setting.

Adaptation of Estimators in Nonstationary Environments An important theoretical difference between the estimators arises in nonstationary environments. The sample-average estimator uses a step size of $1/N_t(a)$ in its update rule

$$Q_{t+1}(a) = Q_t(a) + \frac{1}{N_t(a)} \left(R_t - Q_t(a)\right),$$

which decreases as the number of observations increases. Over time, this causes the estimator to become increasingly resistant to new information. If the true action values drift, the estimator changes only very slowly because recent rewards receive very little weight.

In contrast, estimators that use a constant step size maintain a fixed learning rate

$$Q_{t+1}(a) = Q_t(a) + \alpha \left(R_t - Q_t(a)\right),$$

which ensures that recent observations continue to influence the value estimate. Sliding-window estimators follow a similar principle by explicitly discarding older rewards and computing the average over only the most recent observations. As a result, these estimators can track changes in the reward distribution more effectively when the environment is nonstationary. This theoretical distinction explains the empirical behaviour observed in the experiments, where constant step-size and sliding-window estimators outperform the sample-average estimator in the nonstationary bandit environment.

3.2 Stationary Environment Results

Figures 6 and 7 show the stationary learning curves. All estimators perform reasonably well, and the differences between methods are relatively small compared with the nonstationary setting. This behaviour is expected because in a stationary environment past observations remain valid, allowing estimators such as the sample-average method to converge reliably toward the true action values.

Sliding window achieves the highest final optimal-action rate, while sample average remains highly competitive in terms of average reward. The unbiased constant step-size estimator also performs well, combining stable learning with reduced initialization bias. In contrast, the constant step-size estimator performs slightly worse in this stationary setting because it continues to place significant weight on recent noisy rewards rather than averaging all observations equally.

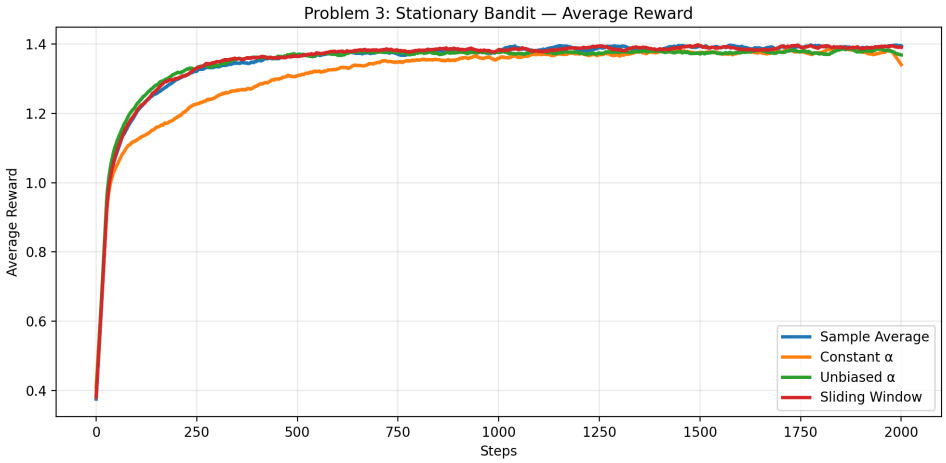

Figure 6: Problem 3: Average reward for action-value estimators in the stationary bandit.

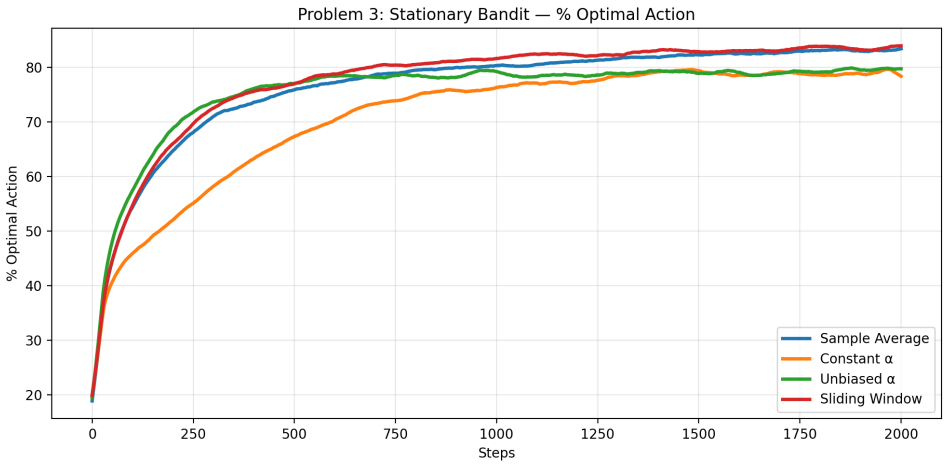

Figure 7: Problem 3: Percentage of optimal actions for action-value estimators in the stationary bandit.

Table 4: Problem 3 stationary summary.

Estimator	Final Avg Reward	AUC Avg Reward	Final % Optimal	AUC % Optimal
Sample Average	1.3984	1.3538	83.70	76.62
Constant α	1.3027	1.3201	77.35	71.00
Unbiased α	1.3658	1.3508	79.85	75.73
Sliding Window	1.3808	1.3561	84.00	77.69

3.3 Nonstationary Environment Results

Figures 8 and 9 show the results for the nonstationary bandit environment. In contrast to the stationary case, the differences between estimators become more pronounced when the true action values change over time.

The sample-average estimator performs worst in terms of optimal-action selection because its effective learning rate decreases as more observations are collected. As a result, it becomes increasingly slow to adapt when the underlying action values drift.

Estimators that place greater emphasis on recent rewards perform better in this setting. The constant step-size estimator improves adaptability by maintaining a fixed learning rate, allowing value estimates to track changing reward distributions. The unbiased constant step-size estimator further improves performance by reducing the bias introduced by initial value estimates while retaining the adaptability of constant step-size updates.

The sliding-window estimator also performs strongly because it computes value estimates using only the most recent observations, allowing it to respond quickly to changes in the underlying action values. As a result, adaptive estimators such as unbiased constant step-size and sliding-window methods achieve the strongest overall performance in the nonstationary environment.

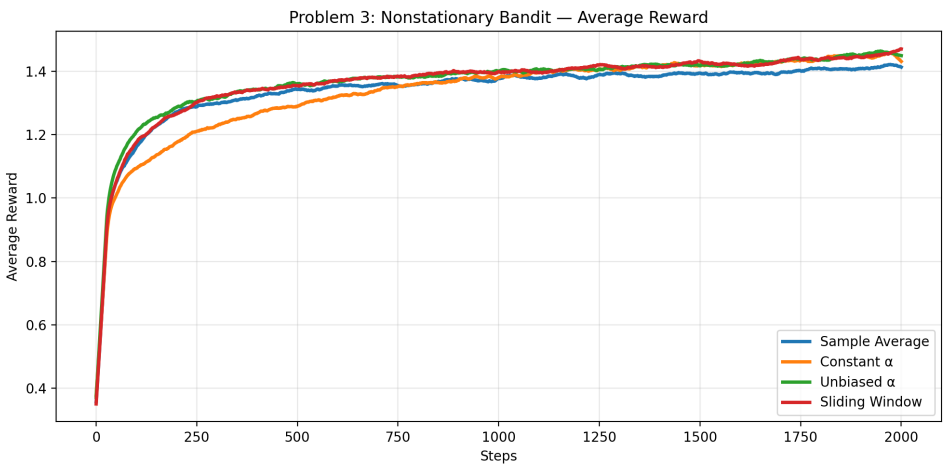

Figure 8: Problem 3: Average reward for action-value estimators in the nonstationary bandit.

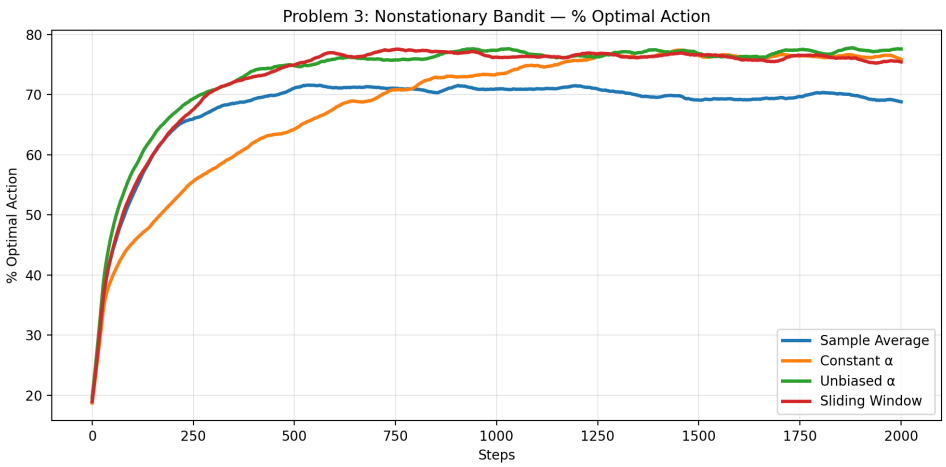

Figure 9: Problem 3: Percentage of optimal actions for action-value estimators in the nonstationary bandit.

Table 5: Problem 3 nonstationary summary.

Estimator	Final Avg Reward	AUC Avg Reward	Final % Optimal	AUC % Optimal
Sample Average	1.4077	1.3439	68.50	68.04
Constant α	1.4063	1.3382	75.20	68.84
Unbiased α	1.4474	1.3707	77.55	73.67
Sliding Window	1.4733	1.3673	75.20	73.08

3.4 Discussion

Problem 3 demonstrates that the choice of action-value estimator becomes particularly important when the environment is nonstationary. In stationary environments, several estimators achieve similar performance over a finite horizon because the true action values remain fixed. Under these conditions, the sample-average estimator performs well because it converges to the expected reward of each action as more observations are collected.

In contrast, the nonstationary environment introduces drifting action values, causing the optimal action to change over time. In this setting, the sample-average estimator performs poorly because its effective learning rate decreases with the number of observations. As a result, new rewards have little influence on the estimate, making the algorithm slow to adapt to changes in the underlying reward distribution.

Estimators that maintain a fixed learning rate, such as constant step-size methods, adapt more effectively because they place greater weight on recent observations. The unbiased constant step-size estimator further improves this behaviour by reducing the bias introduced during early learning. The sliding-window estimator achieves similar adaptability by discarding older rewards and computing value estimates using only recent observations.

These results illustrate an important principle in reinforcement learning: estimators that average over the entire reward history are well suited to stationary environments, while estimators that emphasise recent observations are better suited to nonstationary environments.

Conclusion

This report examined the multi-armed bandit problem from three complementary perspectives. Problem 1 showed that exploration is essential: greedy policies become trapped by noisy early estimates, while ϵ-greedy policies achieve substantially better long-term performance. Problem 2 showed that not all exploration strategies are equally effective; UCB consistently produced the strongest results in the stationary testbed because it directs exploration toward uncertain actions. Problem 3 demonstrated that the choice of action-value estimator becomes particularly important in nonstationary environments, where adaptive estimators such as unbiased constant step-size and sliding-window methods outperform sample-average estimation. Overall, the experiments confirm that strong bandit performance depends on both effective exploration strategies and appropriate value estimation, particularly when environments evolve over time.

References

[1] Sutton, R. S., and Barto, A. G. *Reinforcement Learning: An Introduction.* 2nd ed., MIT Press, 2018.

[2] Auer, P., Cesa-Bianchi, N., and Fischer, P. Finite-time analysis of the multiarmed bandit problem. *Machine Learning*, 47(2–3), 235–256, 2002.

[3] Lai, T. L., and Robbins, H. Asymptotically efficient adaptive allocation rules. *Advances in Applied Mathematics*, 6(1), 4–22, 1985.

[4] Bubeck, S., and Cesa-Bianchi, N. Regret analysis of stochastic and nonstochastic multi-armed bandit problems. *Foundations and Trends in Machine Learning*, 5(1), 1–122, 2012.

[5] Szepesvári, C. *Algorithms for Reinforcement Learning*. Morgan and Claypool Publishers, 2010.